

Generative AI Fundamentals

1. Architectural Core & Value Proposition

Definition & Overview

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

Why? (Problem Solved)

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

How it Solves the Problem

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and

ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

Key Benefits

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these

technologies are widely adopted and how they support enterprise application development.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

2. Foundational Pillars & Key Concepts

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning

patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer

architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

3. Core Concepts

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages,

limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

4. Engineering Tooling & Implementation

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and

ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering,

hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

5. Operational Best Practices, Risks & Security

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages,

limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows,

implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

6. Key Takeaways

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering,

hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.

Based on this module, I understand that Generative AI focuses on creating new content rather than only analyzing existing data. Large Language Models built using transformer architectures can generate text, code, images, and other forms of content by learning

patterns from massive datasets. I also learned about prompt engineering, hallucinations, retrieval augmented generation (RAG), fine-tuning, responsible AI, and ethical considerations. These concepts helped me understand both the capabilities and limitations of modern AI systems. I also explored related tools, workflows, implementation approaches, best practices, security considerations, advantages, limitations, and industry use cases. This strengthened my confidence in explaining the topic during assessments and interviews.